

Assignment – 09:

Problem Statement:

Write a program in C to check whether the given number is an Armstrong number or not.

An Armstrong number is an integer with three digits such that the sum of the cubes of its digits is equal to the number itself.

Example: 371

Code:

```
#include <stdio.h>

int main()
{
    int num,temp,digit,sum=0;
    printf("Enter a 3-Digit number: ");
    scanf("%d",&num);
    temp=num;
    while (temp!=0)
    {
        digit=temp%10;
        sum=sum+(digit*digit*digit);
        temp=temp/10;
    }
    if (sum==num)
    {
        printf("%d is an Armstrong Number",num);
    }
    else
    {
        printf("%d is not an Armstrong Number",num);
    }
}
```

}

Input & Output:

```
Enter a 3-Digit number: 153
153 is an Armstrong Number
```

```
Enter a 3-Digit number: 103
103 is not an Armstrong Number
```

```
Enter a 3-Digit number: 370
370 is an Armstrong Number
```

